AstroPixel: Mission Designer: pitch script

Team Shay Rafe3 · NASA Space Apps 2026 · Space Mission Design Game · Ramallah · 5-minute slot: 3:40 presentation (3:00 spoken + 0:40 video), then Q&A

The story in one line (everyone memorises it):
"Millions love space, but nothing lets them run a real mission. AstroPixel lets anyone, anywhere, build and fly a whole mission with real NASA numbers, and learn why every part depends on every other."

1. Timing plan

#	Slide	Time	Length	Criteria	Speaker
1	Title + hook	0:00–0:20	20 s	Presentation	
2	The problem: millions love space, no tools to see a mission	0:20–1:00	40 s	Impact	
3	Close to home: Palestine	1:00–1:30	30 s	Impact	
4	Solution: AstroPixel	1:30–2:05	35 s	Creativity, Relevance	
5	Science + NASA data	2:05–2:35	30 s	Relevance, Validity	
6	Demo video (5 s intro + 40 s video)	2:35–3:20	45 s	Validity, Presentation	demo driver
7	Impact, next steps, team + close	3:20–3:40	20 s	Impact, Validity	

The presentation takes **3:40** (3:00 spoken + 0:40 video). That leaves about 1:20 of the 5-minute slot for questions. At a calm pace (about 150 words a minute) the spoken parts take about 2:55. Rehearse with a timer and stop at 3:40.

2. The script, slide by slide

Slide 1: Title + hook `0:00–0:20`

> [HOOK, about 12 s: pick one from section 4]
>
> "We're Team Shay Rafe3 from Ramallah, and this is **AstroPixel: Mission Designer**, our answer to the Space Mission Design Game challenge."

Slide 2: The problem `0:20–1:00`

> "Space has never been more popular. NASA's Artemis II coverage this year got 149 million views. 222 million people follow NASA online. And more than 114,000 people joined Space Apps last year.

> But what can all those people actually do with that passion?
>
> They can **watch** a launch, but not the hundreds of decisions behind it. They can **study** one topic at a time: orbits, then power, then radios. Or they can **play** a space game with no budget, no power limits, no data and no team.
>
> **Nothing lets you run a real mission yourself, from the first idea to the final result.**"

Point at the three numbers, then at WATCH / READ / PLAY. Pause after "with that passion?"

Slide 3: Close to home `1:00–1:30`

> "Here in Palestine, that gap is even wider. We found no aerospace engineering degree at our universities. The best-known way into NASA is a trip for a handful of teenagers. And hands-on space learning here mostly means small volunteer astronomy clubs.
>
> Young Palestinians are as curious about space as anyone.
>
> **The knowledge exists. The access doesn't.**"

Check before the pitch: the three Palestine facts (section 6). If you can't confirm one, soften it ("very few", "limited").

Slide 4: Our solution `1:30–2:05`

> "So we built AstroPixel: a whole NASA-style mission, in your browser.
>
> You pick a destination and build your spacecraft from 29 real parts while Mission Control tracks every resource live. You launch, cruise through real solar storms and land in real gravity. When something fails, the game tells you exactly why, so you fix it and fly again, alone or as a team of four roles who must all agree.
>
> And players discover what real engineers know: **add one instrument, and you need more power, more mass, more fuel, a bigger rocket and more money.**"

That last line is the challenge's core idea (every instrument affects every other part). Say it slowly while pointing along the bottom banner.

Slide 5: The science + NASA data `2:05–2:35`

> "Underneath, it's real science. The rocket equation: every extra kilogram needs more fuel. Sunlight and radio both fade with distance squared, so at Saturn, with one percent of Earth's sunlight, you need nuclear power and a big dish. NASA JPL's design margins decide when a system is safe.
>
> And the space storms are real records from NASA's DONKI database, like the extreme storm of May 2024. Every number has a source."

The speaker for this slide studies section 3 so they can answer follow-up questions.

Slide 6: Demo video `2:35–3:20`

> "Let's see it fly." *[play the 40 s video and stay quiet]*

What the video should show, in this order: the builder (one part changes the lamps) → launch → a DONKI storm card in cruise → landing → the report's cause chain ("why it failed") → fix one part → success. English subtitles are required.

Slide 7: Impact, next steps, team `3:20–3:40`

> "AstroPixel is free and runs in any browser: no install, no account, no travel. Next, we'll test it with a school class here in Ramallah and add real NASA images.
>
> **Lose your first spacecraft here, for free, and learn why.** Thank you."

3. Science study sheet (for slide 5 and the Q&A)

Every number below is in `game/scripts/data.gd` with its source. Learn the plain-words line first, then the example.

3.1 The rocket equation: why mass snowballs

$$\Delta v = Isp \cdot g_0 \cdot \ln(m_0 / m_1)$$

- **Plain words:** to change speed by Δv, the ship has to throw fuel out the back. The heavier the ship, the more fuel it needs, and that fuel is extra weight too.
- **Symbols:** Δv = change in speed (m/s) · Isp = engine efficiency in seconds (game: **300 s**, NASA GMAT default) · g_0 = 9.81 m/s² · m_0 = mass with fuel · m_1 = mass after the burn.
- **Example (Mars arrival, Δv = 2.1 km/s):** m_0/m_1 = e^(2100 / (300 × 9.81)) = e^0.714 ≈ **2.04**. So about **half the ship's starting mass is fuel**. A 1,000 kg ship needs about 1,040 kg of fuel. Add a 100 kg instrument and you need about **104 kg more fuel**.
- **In the game:** each part's mass is fed into this, and the rocket's lift capacity is a hard limit with a 35% margin.

3.2 Sunlight falls with distance squared: why Saturn needs nuclear power

$$S = 1361 / d^2 \ (W/m^2, \ d \text{ in AU})$$

- **Plain words:** sunlight spreads over a bigger sphere the farther you go. Twice as far means a quarter of the light.
- **Real values (NSSDCA Planetary Fact Sheets):** Earth 1,361 W/m² · Mars (1.52 AU) 586 W/m², 43% · Jupiter (5.2 AU) 50 W/m², under 4% · Saturn (9.54 AU) **14.8 W/m², about 1%**.
- **Example:** a 1 kW solar array near Earth gives only about 11 W at Saturn. An **MMRTG** nuclear generator gives **110 W, day and night, at any distance**: 45 kg, 14-year design life (NASA MMRTG fact sheet). Perseverance and Curiosity use one.
- **Also real:** solar arrays about 30 W/kg (up to 200), Li-ion batteries 150–270 Wh/kg (NASA Small Spacecraft Technology State of the Art).

3.3 Radio: distance squared again, plus delay

$$\text{signal} \propto 1 / d^2 \qquad \text{delay} = \text{distance} / \text{speed of light}$$

- **Plain words:** the radio signal weakens the same way sunlight does. Far away you need a bigger dish, more power, or you send less data.
- **Example:** at Saturn the signal is about 1/90 as strong as from 1 AU ($9.5^2 \approx 90$). Light takes 8.3 minutes per AU, so Saturn's delay is **about 80 minutes each way**. Mars is 4–24 minutes; the Moon 1.3 seconds.
- **Why landers land themselves:** with a 4 to 24 minute delay, nobody on Earth can steer a Mars landing live.
- **Real dishes:** MRO's 3 m high-gain dish sends up to 6 Mbps from Mars; LRO's Ka-band sends 100–300 Mbps from the Moon (NASA NSSDCA).

3.4 Margins: why "just enough" fails

- **Plain words:** early designs are always wrong by a bit, so NASA keeps spare capacity on purpose.
- **Real values (JPL Concept Maturity Level 4 checklist, Wessen et al. 2013): 35%** margin on dry mass, power and energy; **3 dB** on the radio link. 3 dB means twice the signal power ($10 \cdot \log_{10} 2 \approx 3$). Budget: **25% cost reserve** (NASA Discovery 2019 call).

- **In the game:** a Mission Control lamp turns green only when the margin is kept. Random events eat margin, so a ship with no margin fails at the first surprise.

3.5 Real space weather: NASA DONKI

- **What it is:** DONKI = Space Weather Database Of Notifications, Knowledge, Information (NASA, via api.nasa.gov). We use its **GST (geomagnetic storm)** records from 2024.
- **Kp index:** 0 (calm) to 9 (extreme). **10 May 2024, Kp 9** was the strongest storm in about 20 years; auroras were seen very far south.
- **In the game:** a real storm hits the ship in cruise. Without radiation shielding a part can fail; with shielding it's a non-event. The lesson: redundancy and protection turn disasters into nothing.

3.6 Landing physics (if asked)

- **Gravity (NSSDCA):** Earth 9.81 · Moon 1.62 · Mars 3.73 · Venus 8.87 m/s².
- **Mars:** thin air (under 1% of Earth's): needs a heat shield *and* a parachute. Perseverance's chute opened near 1,512 km/h and slowed it to about 306 km/h.
- **Moon:** no air, so parachutes and heat shields are useless: only engines.
- **Venus:** 92 times Earth's pressure, 464 °C. Venera 13 survived 127 minutes on the surface (1982).

3.7 What's real and what's a game rule (be honest when asked)

Real, with a NASA or mission source	Game rules or estimates (labelled in the code)
Gravity, sunlight, distance, light delay · instrument mass/power/data (MRO CTX, CRISM, Cassini MAG) · MMRTG · battery and solar W/kg · MRO/LRO data rates · margins · DONKI storms · Isp 300 s · rocket capacities and prices (company figures)	Mission budgets · some part prices · small antenna rates · the drone · difficulty multipliers · flight times taken from real missions instead of full GMAT runs

4. Hook ideas (team picks one, about 10 s)

1. **The failed launch (visual).** Play the 11 s clip `pitch/intro_rocket_fail.mp4` with no sound, then: *"That rocket failed because of one decision. In real life that mistake costs years and millions. In our game it costs nothing, and it tells you why."*
2. **Put the judge in the chair.** *"You have one shot at Saturn and a fixed budget. A better camera, or a bigger battery? Choose wrong and your spacecraft goes dark 1.4 billion kilometres from home."*
3. **The real disaster.** *"In 1999 NASA lost the Mars Climate Orbiter because one team worked in pounds and another in newtons. One mismatch between two teams. A $327 million spacecraft, gone."* This links directly to "every part depends on every other" and to teamwork. **Verify the number before using it.**
4. **The 80-minute silence.** *"Imagine sending a command and waiting 80 minutes to hear if it worked. That's Saturn. Every decision has to be right before launch."*
5. **Show of hands.** *"Who here has ever dreamed of working on a space mission? Keep your hand up if you know how much power a camera needs at Saturn."* (Hands drop.) *"That's the gap we're closing."*

5. Judges' questions: 3-beat answers (answer → evidence → impact, 20 s max)

Question	Answer
Impact: Who will use it, and how will it reach them?	Students and space clubs first, through teachers. It's one free web link, with no install and no account, so a school computer lab is enough. Next step: pilot with one Ramallah class.
Creativity: What exists already?	Kerbal Space Program is fun, but it has no budget, power, data or comms trade-offs and no NASA sourcing. NASA's own games cover one mission, solo. Ours is the whole mission chain, up to 4 roles, and it explains every failure.
Validity: How do you know it's right?	Every number lives in one file with its source, and invented values are labelled "game rule". An automated self-play test runs all 24 missions. Honest limit: flight times come from real missions, not full GMAT runs.
Relevance: Which NASA data, and what breaks without it?	DONKI GST storms, NSSDCA Planetary Fact Sheets, the MMRTG fact sheet, the Small Spacecraft Technology report, MRO/LRO/Cassini instrument data, the GMAT Isp, JPL CML margins. Without them the physics would be made up: Saturn would not need nuclear power.
Presentation: One sentence?	"Build a spacecraft, fly the whole mission, and learn why every part depends on every other."
Is it only for experts?	No. EASY mode has a Mission Control guide and auto-build, and the science is explained in plain words inside the game. HARD gives no hints.
Did you use AI?	Yes, for code help and drafting text. The idea and the game design are ours. No AI images, video or audio: the art comes from our own Blender and Python scripts.

6. Fact-check before the pitch

- **Palestine 1:** "no aerospace engineering degree we could find". Search each university's programme list (An-Najah, Birzeit, IUG, Al-Quds, PTUK). An-Najah or Birzeit may have aerospace-related courses inside mechanical engineering.

- **Palestine 2:** the Smithsonian National Air and Space Museum × Al Nayzak programme takes Palestinian teens (14–17) to NASA Goddard (global.si.edu).
- **Palestine 3:** hands-on space learning is mostly small volunteer astronomy clubs (e.g. NOVA for Astronomy and Space Science in Ramallah helps universities start clubs; Al-Monitor, 2019). Name any bigger programme you know of, or soften it.
- **Artemis II:** "149 million views" is NASA's own total for its Artemis II coverage (March–April 2026), reported by Space.com as NASA's streaming record. It counts views across all agency platforms, not unique people.
- **NASA followers:** 222 million across platforms is NASA's official 2023 figure (newer official totals weren't found).
- **Space Apps 2025:** 114,000+ participants, 167 countries (NASA figures, quoted in the BZU pitch training).
- **Live demo:** open the web game and the relay server about 2 minutes before you pitch (the free server needs about a minute to wake up). Keep the video offline on the laptop and on a USB stick.

Sources: Space.com: Artemis II breaks NASA's streaming record · NASA: social media · Palestine facts: Smithsonian Global: Connecting Palestinian students with STEM · EduRank: aerospace in Palestine · Astrobites: astronomy in Palestine · UN: UNESCO Chair in Astronomy at IUG.